Paying to Sharpen Identification: Optimal Probing of Coarse Data

Patrick Ferguson

Cornell University

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pcf44@cornell.edu patrickcferguson.github.io

Paying to Improve Data

Data are often rounded, mismeasured, or incomplete.

coarse

Follow-up questions, remeasurement, or record linkage can improve their quality.

probes

Imagine that we could pay to improve our data.

With a limited budget, which coarse observations should we probe?

Heitjan and Rubin (1991).

Example: Probing Bracketed Incomes

Suppose incomes are observed only in brackets.

A probe may:

- reveal the true income
- return a narrower bracket
- provide no further information

With a limited budget, which bracketed incomes should we probe?

Consider:

- initial bracket width
- probe effectiveness
- probe cost

Data Quality and Identification Uncertainty

- Target parameter: mean income θ
- Identified set Θ: all parameter values θ consistent with the brackets and maintained assumptions
- Income (\$1,000s): 50% in $[20, 40]$, 50% in $[60, 100]$

$$\Theta = \left[\frac{20+60}{2}, \frac{40+100}{2}\right] = [40, 70], \quad \operatorname{diam}\Theta = 30$$

Downstream decisions

Two Ways to Sharpen Identification

Two ways to shrink the identified set for mean income:

- Stronger assumptions: each bracket's mean income = its midpoint
- Better data: narrower income brackets from follow-up questions

This paper: the second approach

Probe data to minimise identification uncertainty.

What This Paper Contributes

1. Optimal rule under a limited budget: probe in decreasing order of

$$\frac{\text{expected diameter reduction}}{\text{marginal cost}}$$

2. Same scalar outcome: one optimal rule for its mean and full distribution when the distance between distributions is measured by the area between their CDFs.

Related Literature

- Manski and Molinari (2008): trade-off between survey cost and data quality
- Dominitz and Manski (2017, 2022); Tetenov (2012): budget allocation between more observations and better data

This paper: allocate budget across observations to maximise data quality

Full literature

Outline

1. Coarse Data and Partial Identification
2. Optimal Probing
3. Distributional Targets
4. Survey-Design Application 1: Subjective Job Insecurity
5. Survey-Design Application 2: Stock-Return Expectations
6. Extensions, Limits, and Conclusion

Coarse Data and Partial Identification

From Coarse Data to Partial Identification

- Observed: $\mathbf{z}_0 \sim \mathrm{P}$; P known (focus on identification)
- Outcome restriction: $\mathbf{y} \in Y(\mathbf{z}_0)$
- Q: joint distribution of $\mathbf{z}_0$ and latent $\mathbf{y}$

Compatible joint distributions:

$$\mathcal{Q} := \{\mathrm{Q} \mid \mathrm{Q}_{\mathbf{z}_0} = \mathrm{P},\ \mathrm{Q}(\mathbf{y} \in Y(\mathbf{z}_0)) = 1\}$$

Identified set for the mean of a scalar transformation g:

$$\Theta := \{\mathrm{E}_{\mathrm{Q}}[g(\mathbf{z}_0, \mathbf{y})] \mid \mathrm{Q} \in \mathcal{Q}\}$$

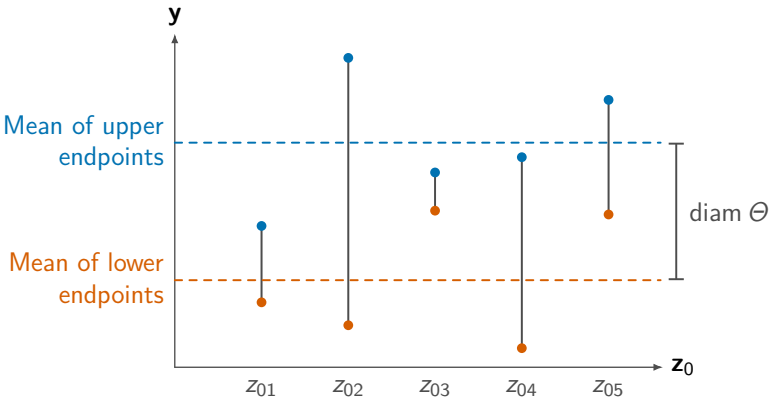
Proposition 1: Scalar Expectation Diameter

- Unit-by-unit restrictions
- Bounded measurable scalar transformation g

Then

$$\operatorname{diam}\Theta = \mathrm{E}[\Delta_0(\mathbf{z}_0)], \quad \Delta_0(z_0) := \operatorname{diam}\{g(z_0, y) \mid y \in Y(z_0)\}$$

Illustrating Proposition 1



$g(z_0, y) = y$; five equally weighted observations

Proposition 1: Proof Sketch

Pointwise lower and upper envelopes:

$$g^L(z_0) := \inf_{y \in Y(z_0)} g(z_0, y), \quad g^U(z_0) := \sup_{y \in Y(z_0)} g(z_0, y)$$

Unit-by-unit restrictions $\implies$ sharp mean bounds by averaging the pointwise envelopes

$$\operatorname{diam} \Theta = \mathrm{E}[g^U(\mathbf{z}_0)] - \mathrm{E}[g^L(\mathbf{z}_0)] = \mathrm{E}[\Delta_0(\mathbf{z}_0)]$$

▸ Sharpness details

Other Scalar Targets

Target	Transformation $g(z_0, y)$
Mean	y
CDF value at t	$\mathbf{1}\{y \leqslant t\}$
Conditional mean within C	$y\mathbf{1}\{z_0 \in C\}/\mathrm{P}(C)$

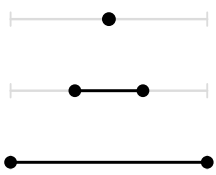
Optimal Probing

Probing Refines the Outcome Set

Probe $\longrightarrow$ additional information $\mathbf{z}_1 \longrightarrow$ refined outcome set:

$$Y(z_0, \mathbf{z}_1) \subseteq Y(z_0)$$

- Exact successful measurement: singleton
- Incomplete measurement: a smaller set
- Failed probe: the original set



A Probing Rule and Its Identified Set

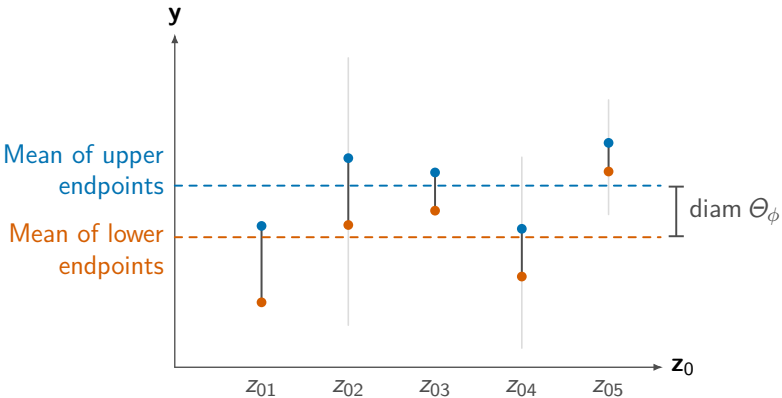
Binary probing rule: $\phi : \mathcal{Z}_0 \to \{0, 1\}$ induces

$$Y_\phi(z_0, z_1) := \begin{cases} Y(z_0, z_1), & \phi(z_0) = 1 \\ Y(z_0), & \phi(z_0) = 0 \end{cases}$$

and identified set Θ_ϕ: all values of $\mathrm{E}[g(\mathbf{z}_0, \mathbf{y})]$ consistent with

- $y \in Y_\phi(z_0, z_1)$
- observed distribution of $\mathbf{z}_0$ and $\mathbf{z}_1$

How Probing Sharpens Identification



$g(z_0, y) = y$; five equally weighted observations. Probing units $2, 4, 5$

Costly Probing

Probing cost at z_0: $c(z_0) > 0$

Choose which observations to probe:

$$\min_{\phi} \ \operatorname{diam} \Theta_{\phi} \quad \text{subject to} \quad \mathrm{E}[c\phi] \leqslant b$$

The Value of a Probe

$\Delta_1(z_0, z_1)$: diameter of g over the refined outcome set $Y(z_0, z_1)$

Marginal value (expected diameter reduction):

$$v(z_0) := \Delta_0(z_0) - \underbrace{\mathrm{E}[\Delta_1(\mathbf{z}_0, \mathbf{z}_1) \mid \mathbf{z}_0 = z_0]}_{\overline{\Delta}_1(z_0)}$$

- Exact resolution: $v(z_0) = \Delta_0(z_0)$
- Partial refinement by fraction $\alpha(z_0)$: $v(z_0) = \alpha(z_0)\Delta_0(z_0)$
- Resolution with probability $p(z_0)$: $v(z_0) = p(z_0)\Delta_0(z_0)$

Initially: expected reductions treated as known

Theorem 1: Optimal Probing

Suppose P is nonatomic, costs are additive, and each probe refines only its own unit's outcome set. An optimal binary rule satisfies

$$\phi^*(z_0) = \begin{cases} 1, & v(z_0)/c(z_0) > \lambda^* \\ 0, & v(z_0)/c(z_0) < \lambda^* \end{cases}$$

Choose the cutoff to satisfy the budget.

Probe in decreasing order of expected diameter reduction per unit cost.

With atoms, a feasible binary rule has excess diameter at most $\sup_{z_0} \mathrm{P}(\{z_0\})v(z_0)$ relative to the unrestricted optimum.

Estimated values

Theorem 1: Proof Sketch

1. Post-probing diameter (Proposition 1)

$$\operatorname{diam} \Theta_\phi = \mathrm{E}[\Delta_0] - \mathrm{E}[v\phi]$$

2. Rewrite problem:

$$\max_{\phi} \ \mathrm{E}[v\phi] \quad \text{subject to} \quad \mathrm{E}[c\phi] \leqslant b$$

3. Neyman–Pearson solution

$$\text{probe when } \frac{v(z_0)}{c(z_0)} > \lambda^*$$

Cutoff: fit the budget. Ties: binary selection permitted by nonatomicity.

Lehmann and Romano (2022), Theorem 3.2.1.

Distributional Targets

Wasserstein-1 Distance and Diameter

- Target: **full distribution** of $g(\mathbf{z}_0, \mathbf{y})$; identified set Θ_{dis}
- Univariate Wasserstein-1 distance (area between CDFs):

$$W_1(\mu, \nu) = \int_{\mathbb{R}} |F_\mu(t) - F_\nu(t)| \, \mathrm{d}t, \quad F_\mu(t) := \mu\big(g(\mathbf{z}_0, \mathbf{y}) \leqslant t\big)$$

- Identified-set diameter:

$$\operatorname{diam}_{W_1} \Theta_{\text{dis}} := \sup_{\mu, \nu \in \Theta_{\text{dis}}} W_1(\mu, \nu)$$

Villani (2009), Definition 6.1; Panaretos and Zemel (2019), Corollary 1.5.3.

Proposition 3: Wasserstein-1 Diameter

Same unit-by-unit restrictions and scalar transformation g:

$$\operatorname{diam}_{W_1} \Theta_{\text{dis}} = \mathrm{E}[\Delta_0] = \operatorname{diam} \Theta$$

For fixed g, the same rule minimises diameter for both the mean and the full distribution.

▸ CDF illustration ▸ Total variation ▸ CDF L^∞

Proposition 3: Proof Sketch (Lower Bound)

Case $g(z_0, y) = y$; $\operatorname{diam}_{W_1} \mathcal{Q}$ uses y-marginals

Kantorovich–Rubinstein duality:

$$W_1(\mathrm{Q}_y, \mathrm{Q}'_y) = \sup_{f \in \mathrm{Lip}_1(\mathbb{R})} \left| \mathrm{E}_{\mathrm{Q}}[f(\mathbf{y})] - \mathrm{E}_{\mathrm{Q}'}[f(\mathbf{y})] \right|$$

Take $f(t) = t$:

$$W_1(\mathrm{Q}_y, \mathrm{Q}'_y) \geqslant \left| \mathrm{E}_{\mathrm{Q}}[\mathbf{y}] - \mathrm{E}_{\mathrm{Q}'}[\mathbf{y}] \right|$$

Proposition 1:

$$\operatorname{diam}_{W_1} \mathcal{Q} \geqslant \operatorname{diam}\{\mathrm{E}_{\mathrm{Q}}[\mathbf{y}] \mid \mathrm{Q} \in \mathcal{Q}\} = \mathrm{E}[\operatorname{diam} Y(\mathbf{z}_0)]$$

Villani (2009), Remark 6.5. $\mathrm{Lip}_1(\mathbb{R})$: 1-Lipschitz functions

Proposition 3: Proof Sketch (Upper Bound)

Case $g(z_0, y) = y$; $\operatorname{diam}_{W_1} \mathcal{Q}$ uses y-marginals

Kantorovich–Rubinstein duality:

$$W_1(\mathrm{Q}_y, \mathrm{Q}'_y) = \sup_{f \in \mathrm{Lip}_1(\mathbb{R})} \left| \mathrm{E}_{\mathrm{Q}}[f(\mathbf{y})] - \mathrm{E}_{\mathrm{Q}'}[f(\mathbf{y})] \right|$$

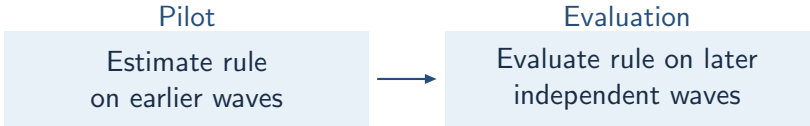
$$\begin{aligned}
\operatorname{diam}_{W_1} \mathcal{Q} &:= \sup_{\mathrm{Q}, \mathrm{Q}' \in \mathcal{Q}} W_1(\mathrm{Q}_y, \mathrm{Q}'_y) \\
&= \sup_{f \in \mathrm{Lip}_1(\mathbb{R})} \operatorname{diam}\{\mathrm{E}_{\mathrm{Q}}[f(\mathbf{y})] \mid \mathrm{Q} \in \mathcal{Q}\} \\
&= \sup_{f \in \mathrm{Lip}_1(\mathbb{R})} \mathrm{E}[\operatorname{diam} f(Y(\mathbf{z}_0))] \leqslant \mathrm{E}[\operatorname{diam} Y(\mathbf{z}_0)] \quad \text{(1-Lipschitz } f\text{)}
\end{aligned}$$

Villani (2009), Remark 6.5. $\mathrm{Lip}_1(\mathbb{R})$: 1-Lipschitz functions

Survey-Design Application 1: Subjective Job Insecurity

Counterfactual Survey-Design Applications

- Job insecurity: initial ambiguity and target weights
  Which observations contribute most to identification uncertainty?
- Stock-return expectations: probe effectiveness
  Which respondents are most likely to provide an informative answer?



Remaining intervals $[y_i^L, y_i^U]$:

$$\widehat{\Theta}_n := \left[\frac{1}{n} \sum_i y_i^L, \frac{1}{n} \sum_i y_i^U \right]$$

Subjective Job Insecurity

Reported subjective chances (%):

Initial question		Follow-up question
loss_i: Job loss within 12 months	$\longrightarrow$	find_i: Finding a job at least as good, conditional on job loss

$$\text{ins}_i := \text{loss}_i \left(1 - \frac{\text{find}_i}{100}\right)$$

Manski and Straub (2000).

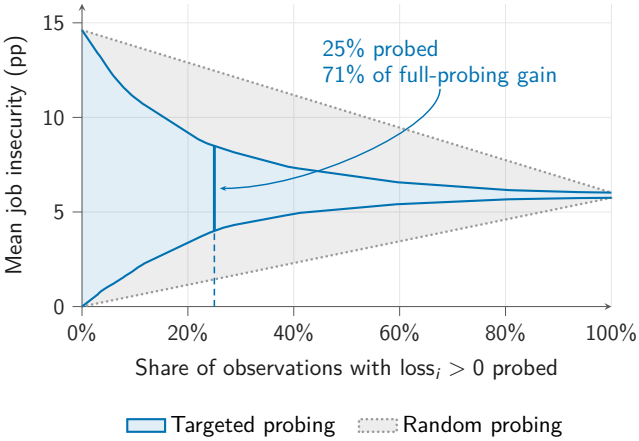
Targeting the Overall Mean or Distribution

$$[0, \text{loss}_i] \xrightarrow{\text{probe}} \begin{cases} \{\text{ins}_i\}, & \text{valid follow-up response} \\ [0, \text{loss}_i], & \text{otherwise} \end{cases}$$

- Uniform probing costs: probe in decreasing order of loss_i
- Pilot: waves 1–8, $n = 3{,}708$
- Evaluation: waves 9–16, $n = 2{,}618$
- Follow-up failures (about 1%): ignored in rule but retained in evaluation

▸ Planned vs realised shares ▸ Loss distribution

At 25% Probed, Targeting Captures 71% of Full Gain



Probe share (%)	Targeted-rule diameter (pp)
0	14.7
25	4.5
100	0.3

▸ Confidence intervals

Targeting a Conditional-Mean Contrast

Manski and Straub: Black–White contrast in mean job insecurity

$$\mathrm{E}[\mathrm{ins}_i \mid i \in B] - \mathrm{E}[\mathrm{ins}_i \mid i \in W] = \mathrm{E}\Bigg[\underbrace{\left(\frac{\mathbf{1}\{i \in B\}}{\mathrm{P}(B)} - \frac{\mathbf{1}\{i \in W\}}{\mathrm{P}(W)}\right)}_{w_i} \mathrm{ins}_i\Bigg]$$

Diameter contribution:

$$\Delta_{0i} = |w_i| \cdot \mathrm{loss}_i$$

Probe in decreasing order of $|w_i^{\mathrm{pilot}}| \cdot \mathrm{loss}_i$

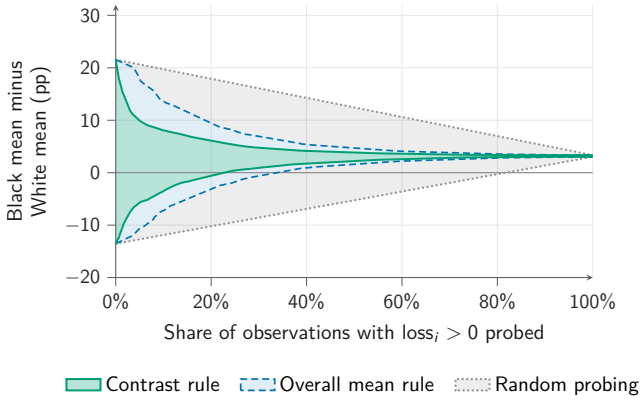
Relative Weights for the Contrast

$$\frac{n_W^{\text{pilot}}}{n_B^{\text{pilot}}} = \frac{3{,}188}{244} \approx 13.1$$



Relative $|w_i^{\text{pilot}}|$: White $= 1$; pilot waves 1–8

Identified Set for Contrast vs Share Probed



Probe share when $0 \notin \widehat{\Theta}$

Rule	Share (%)
Random	82
Overall mean	34
Contrast	23

▸ Diameter ▸ Confidence intervals ▸ Regression coefficient

Survey-Design Application 2: Stock-Return Expectations

Stock-Return Expectations: Predicting Probe Effectiveness

- Target: mean subjective probability that an investment in a diversified stock mutual fund increases in value over the following year, pnr_i
- Common initial outcome set; probe: ask this question

$$[0, 100] \xrightarrow{\text{probe}} \begin{cases} \{\text{pnr}_i\}, & \text{valid response} \\ [0, 100], & \text{otherwise} \end{cases}$$

- Response probability $p(z_0)$:

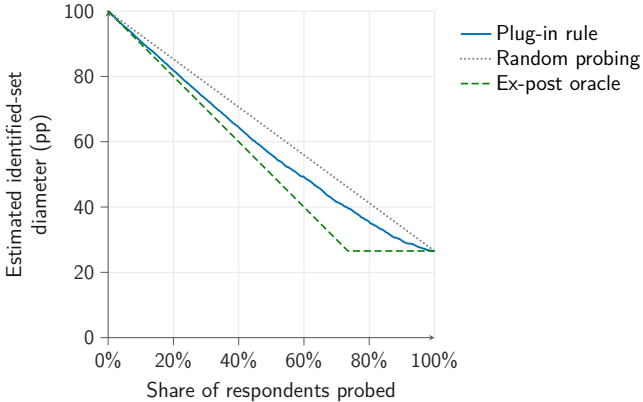
$$v(z_0) = 100\, p(z_0)$$

Learning a Rule from the Pilot

- Ridge-regularised logit with cross-validation $\longrightarrow \widehat{p}(z_0)$
  Plug-in rule: probe in decreasing order of $\widehat{p}(z_0)$
- Predictors: demographics and nonresponse to earlier items
- Pilot: wave 12, $n = 547$
- Evaluation: waves 13–14, $n = 1{,}104$
- Benchmarks: random probing; ex-post oracle (knows who responds)

▸ Plug-in bound ▸ Prediction details

Predicted Effectiveness Improves on Random Probing



Share of full-probing gain at 25% probed

Rule	Share (%)
Random	25
Plug-in	31
Ex-post oracle	34

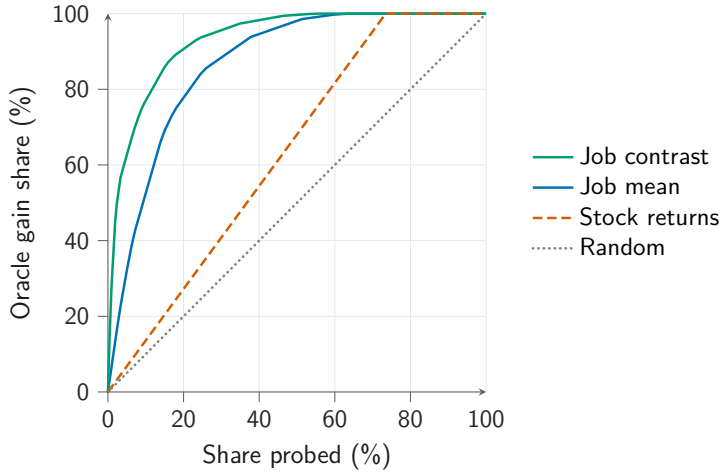
Corollary 1: Inequality and Targeting Gains

Under Theorem 1:

Greater cost-weighted inequality in v/c (Lorenz order)
$\Longrightarrow$ weakly larger optimal gain share at every budget share

▸ Formal statement ▸ Illustration

Targeting Gains across Applications



Full-sample probe shares, including zero-value observations

▸ Loss distribution

Conditional-Mean Diameters by Schooling Level

More nonresponse at lower schooling $\Longrightarrow$ rule targets higher schooling

	Diameter (pp) at 25% probed	
Schooling group	Targeted	Untargeted
No postsecondary education	91	87
Some postsecondary education	82	81
Bachelor's degree or higher	64	79

Can we protect group means while targeting the overall mean?

Proposition 5: Multiple Linear Constraints

Impose $\mathrm{E}[c_k\phi] \leqslant b_k,\ k = 1, \ldots, K$.

A feasible ϕ^* is optimal if some $\lambda_k^* \geqslant 0$ satisfy

$$\phi^*(z_0) = \begin{cases} 1, & v(z_0) > \sum_{k=1}^{K} \lambda_k^* c_k(z_0) \\ 0, & v(z_0) < \sum_{k=1}^{K} \lambda_k^* c_k(z_0) \end{cases},$$

and $\lambda_k^*(b_k - \mathrm{E}[c_k\phi^*]) = 0$ for every k.

Schooling groups L, M, H (pilot sizes n_L, n_M, n_H): equal predicted reductions in conditional-mean diameters

$$\frac{1}{n_L}\sum_{i\in L}\widehat{p}_i\phi_i = \frac{1}{n_M}\sum_{i\in M}\widehat{p}_i\phi_i = \frac{1}{n_H}\sum_{i\in H}\widehat{p}_i\phi_i$$

Reducing Disparities across Schooling Groups

Diameter (pp) at 25% probed

		Targeted	
Target	Untargeted	Unconstrained	Constrained
Overall mean	81.6	77.4	78.1
No postsecondary education	86.8	91.4	81.1
Some postsecondary education	81.1	82.1	78.1
Bachelor's degree or higher	79.0	63.8	76.1

Constraint: overall diameter +0.7 pp; group gap 28 to 5 pp

Extensions, Limits, and Conclusion

Extensions and Limits

Extensions in the paper

- Multiple alternative probe types
- Selected multivariate targets under geometric conditions

Limits of the framework

- Individual quantiles, other probability metrics, and vector targets: probing values need not add across observations
- Restrictions: unit by unit; probes refine only probed units' outcome sets

▸ Quantiles ▸ Other metrics ▸ Vectors ▸ Probe types

Conclusion

1. Scalar expectations and univariate distributions under W_1: identified-set diameter decomposes into a sum of observation-level contributions
2. Optimal rule: probe in decreasing order of expected diameter reduction per unit cost
3. Under a limited budget, the rule captures a much larger share of gains than does untargeted probing.

Backup Slides

Framework and allocation

- Formal identified set
- Mean sharpness details
- Probing and inference
- Estimated probing values
- Why diameter?
- Related literature
- Lorenz representation
- Lorenz illustration
- Multiple probe types

Applications and extensions

- Job confidence intervals
- Contrast confidence intervals
- Regression coefficient
- Response prediction
- Planned vs realised shares
- Loss distribution
- CDF illustration
- Total variation
- Quantiles
- Other metrics
- Vector targets

Proposition 1: Sharpness Details

Pointwise envelopes:

$$g^L(z_0) := \inf_{y \in Y(z_0)} g(z_0, y), \quad g^U(z_0) := \sup_{y \in Y(z_0)} g(z_0, y)$$

Every compatible joint distribution:

$$\mathrm{E}[g^L(\mathbf{z}_0)] \leqslant \mathrm{E}_{\mathrm{Q}}[g(\mathbf{z}_0, \mathbf{y})] \leqslant \mathrm{E}[g^U(\mathbf{z}_0)]$$

Measurable ε-selections approach both bounds, so they are sharp.

Diameter:

$$\begin{aligned} \operatorname{diam} \Theta &= \sup_{\mathrm{Q} \in \mathcal{Q}} \mathrm{E}_{\mathrm{Q}}[g] - \inf_{\mathrm{Q} \in \mathcal{Q}} \mathrm{E}_{\mathrm{Q}}[g] \\ &= \mathrm{E}[g^U(\mathbf{z}_0) - g^L(\mathbf{z}_0)] = \mathrm{E}[\Delta_0(\mathbf{z}_0)] \end{aligned}$$

Diameter Bounds Downstream Minimax Regret

Absolute-loss point estimation: midpoint of Θ; smallest worst-case error $\operatorname{diam}\Theta/2$

Problem j: minimax regret

$$V_j(\Theta) := \inf_a \sup_{\theta\in\Theta}\{\ell_j(a,\theta) - \inf_{a'} \ell_j(a',\theta)\}$$

For downstream family $\mathcal{J}$, if $\ell_j(a,\cdot)$ is K-Lipschitz for every $j \in \mathcal{J}$ and action a, with common K, then

$$\sup_{j\in\mathcal{J}} V_j(\Theta) \leqslant K \operatorname{diam}\Theta$$

For a sufficiently rich $\mathcal{J}$, this inequality binds.

The identified set's location may also matter.

Formal Identified Set

Compatible joint distributions:

$$\mathcal{Q}(Y_0, \mathsf{P}) := \{\mathsf{Q} \mid \mathsf{Q}_{\mathbf{z}_0} = \mathsf{P},\ \mathsf{Q}(\mathbf{y} \in Y(\mathbf{z}_0)) = 1\}$$

Parameter mapping τ:

$$\Theta_\tau(Y_0, \mathsf{P}) := \{\tau(\mathsf{Q}) \mid \mathsf{Q} \in \mathcal{Q}(Y_0, \mathsf{P})\}$$

- Outcome sets: maintained restrictions, possibly from a reporting model
- No cross-unit restrictions on latent outcomes

Probing and Statistical Inference

- Criterion: aggregate remaining ambiguity under maintained observation-level restrictions
- Randomised rules: average diameter over realised probing regions
- Representative probing can support additional design-based inference, including IPW under suitable conditions.
- Optimising inference under those additional restrictions: a broader design problem

Optimality: stated diameter criterion

Estimated Probing Values

Optimal rule using true v: ϕ^*

$$\widehat{\phi} \in \underset{\substack{\phi \\ \mathsf{E}[c\phi] \leqslant b}}{\arg\max}\, \mathsf{E}[\widehat{v}\phi]$$

$$\operatorname{diam}\Theta_{\widehat{\phi}} - \operatorname{diam}\Theta_{\phi^*} \leqslant \mathsf{E}\left[|\widehat{v} - v| \cdot |\phi^* - \widehat{\phi}|\right]$$

Errors only where allocations differ

Correct ranking

$$\mathsf{E}[c\widehat{\phi}] \geqslant \mathsf{E}[c\phi^*], \quad \frac{v}{c} = h\left(\frac{\widehat{v}}{c}\right), \quad h \text{ weakly increasing}$$

$$\implies \operatorname{diam}\Theta_{\widehat{\phi}} = \operatorname{diam}\Theta_{\phi^*}$$

▸ Optimal rule ▸ Prediction ▸ Backup index

Related Literature

- Manski and Molinari (2008): survey follow-up decisions
- Dominitz and Manski (2017, 2022); Tetenov (2012): larger samples versus better data; sampling versus identification uncertainty
- Heitjan and Rubin (1991): conditions for ignorable coarsening
- Partial identification under data coarsening: e.g., Horowitz and Manski (1998); Manski and Tamer (2002); Molinari (2008); Manski and Molinari (2010); Khan, Ponomareva, and Tamer (2011)
- Random-set methods for partial identification: Beresteanu, Molchanov, and Molinari (2012); Molchanov and Molinari (2018)

Stock-Return Response Prediction

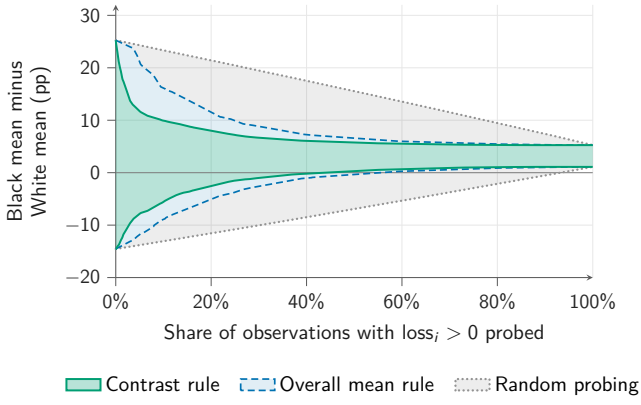
Predictors for ridge-penalised logit

- age, education, sex, race, employment,
  marital status, home ownership, and income
- nonresponse indicators for the earlier weather, crime, health-insurance,
  employment, and income items

Estimation and evaluation

- Penalty: five-fold cross-validation in wave 12; mean squared prediction error
- Fit: full pilot; evaluation: waves 13 and 14, model held fixed
- Probe success includes valid answers implied by survey routing.

Confidence Set for the Contrast



Probe share when $0 \notin \mathcal{C}_n$

Rule	Share (%)
Random	93
Overall mean	56
Contrast	44

$$\widehat{\Theta}_n = [\widehat{\theta}^L, \widehat{\theta}^U], \quad \mathcal{C}_n = \left[\widehat{\theta}^L - 1.96\frac{\widehat{\sigma}^L}{\sqrt{n}}, \widehat{\theta}^U + 1.96\frac{\widehat{\sigma}^U}{\sqrt{n}}\right]$$

Back Backup index

Targeting a Regression Coefficient

Best linear predictor of job insecurity (e_B selects the Black coefficient):

$$\beta_B^{\text{eval}} = \frac{1}{n}\sum_i w_i^{\text{eval}}\,\text{ins}_i, \quad w_i^{\text{eval}} := e_B^{\mathsf{T}}(\widehat{Q}^{\text{eval}})^{-1}x_i, \quad \widehat{Q}^{\text{eval}} := \frac{1}{n}\sum_i x_i x_i^{\mathsf{T}}$$

Probe in decreasing order of

$$|w_i^{\text{pilot}}|\,\text{loss}_i$$

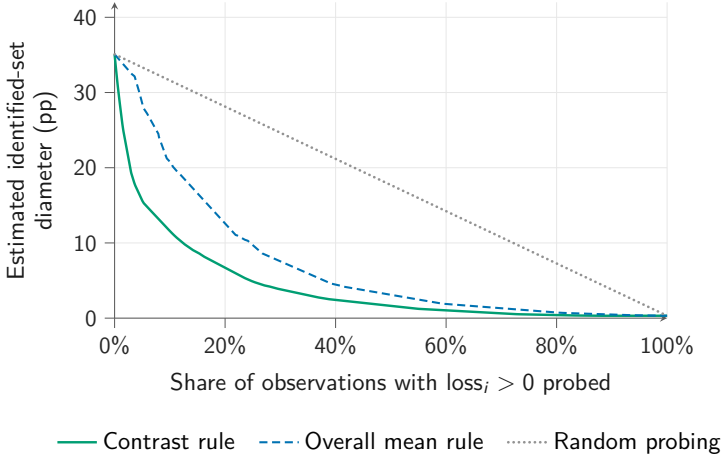
Regressors: age, schooling, sex, race, self-employment, family business; intercept

Relative $|w_i^{\text{pilot}}|$:

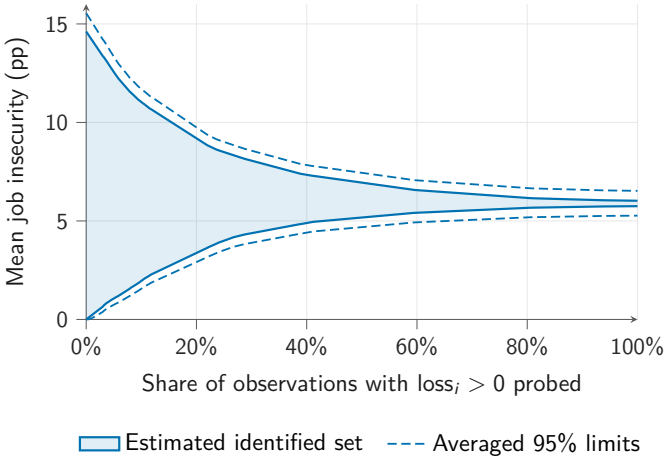


Women aged 35–49; some postsecondary education; employed by others

Estimated Diameter for the Black–White Contrast

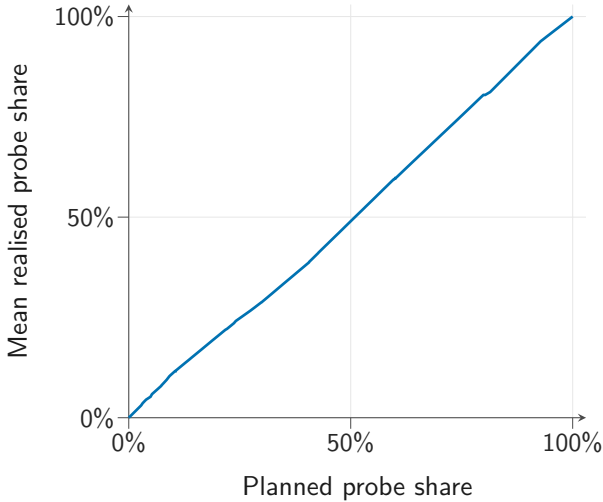


Estimated Sets and Averaged Confidence Limits



▸ Back to results ▸ Backup index

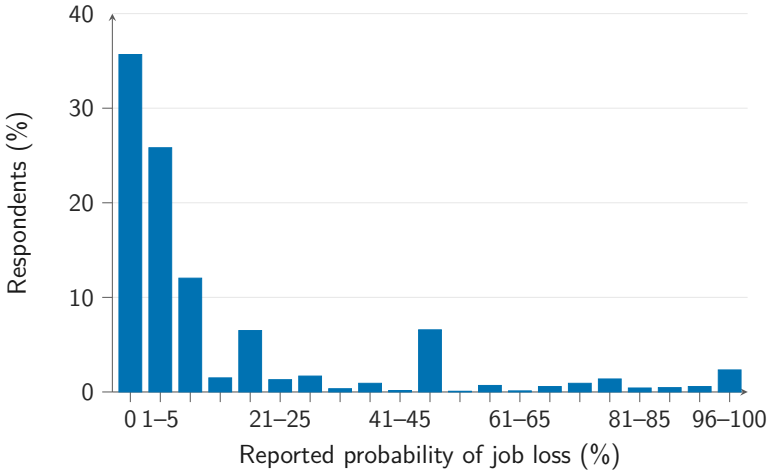
Planned vs Realised Probe Shares



Planned (%)	Realised (%)
10	11.2
25	24.8
50	49.0
75	75.3

Back Backup index

Distribution of Reported Job-Loss Probabilities



5-point bins; evaluation waves 9–16; $n = 2{,}618$

▸ Gains comparison ▸ Job targeting ▸ Backup index

Illustrating Wasserstein-1 Diameter

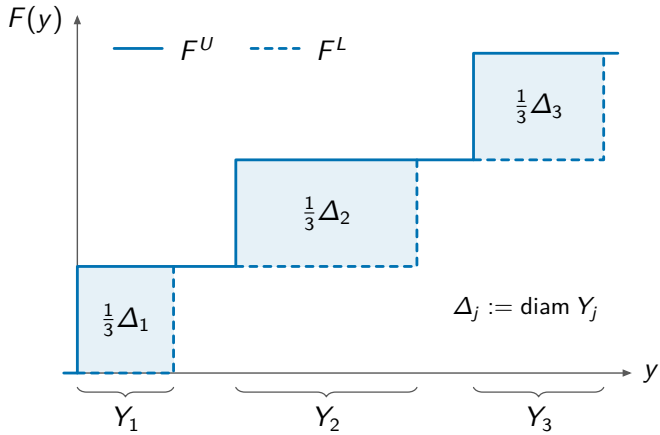
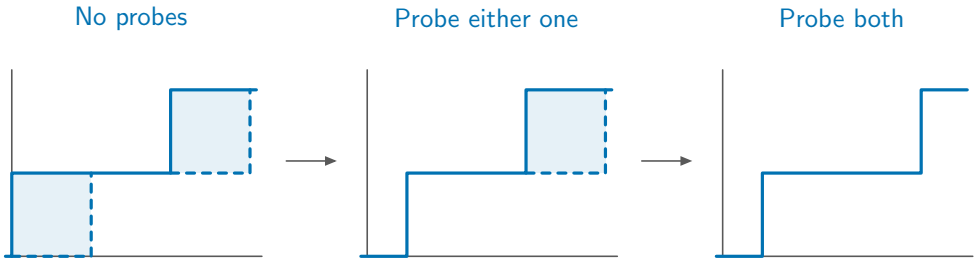


Illustration: $g(z_0, y) = y$; three equally weighted outcome sets

CDF L^∞ Counterexample

$Y_1 = [0, 1], \quad Y_2 = [2, 3]$ Equal weights; exact probes



Nonadditive probing values

Total Variation Diameter Decomposition

TV distance: maximum difference in event probabilities

$$d_{\mathrm{TV}}(\mu, \nu) := \sup_{B \in \mathcal{B}(\mathbb{R})} |\mu(B) - \nu(B)|$$

Suppose there is a Borel set $B \subseteq \mathbb{R}$ such that, for P-almost every z_0, every non-singleton $g(\{z_0\} \times Y(z_0))$ intersects both B and $\mathbb{R} \setminus B$. Then

$$\operatorname{diam}_{d_{\mathrm{TV}}} \Theta_{\mathrm{dis}} = \mathrm{P}\big(\Delta_0(\mathbf{z}_0) > 0\big)$$

Multiple Alternative Probe Types

Action $a \in \{0, \ldots, A\}$ at each z_0; original model: $A = 1$; no probe: $a = 0$

Action a: expected diameter reduction $v_a(z_0)$; cost $c_a(z_0)$

Optimal action: choose from

$$\arg\max_{a=0,\ldots,A} \{v_a(z_0) - \lambda^* c_a(z_0)\}, \quad v_0 = c_0 = 0$$

Plus budget feasibility and complementary slackness

Median Counterexample

Setup:

- 3 observations
- Common outcome set $Y \equiv [0, 1]$
- Probe: corresponding $\mathbf{y}$ revealed exactly

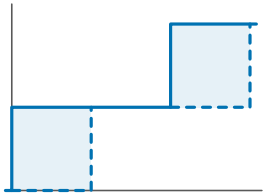
Median m: identified set $[0, 1]$

Probe	Identified Set	Diameter Reduction
Any 1	$[0, 1]$	0
Any 2	$[y_1, y_2]$	$y_1 + 1 - y_2$
All 3	$\{m\}$	1

value of probing non-separable across observations

CDF and Quantile L^p Diameters

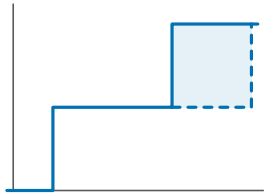
No probes



F: $2^{1/p-1}$

Q: 1

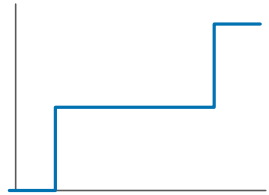
Probe either one



F: $\frac{1}{2}$

Q: $2^{-1/p}$

Probe both



F: 0

Q: 0

Vector Counterexample

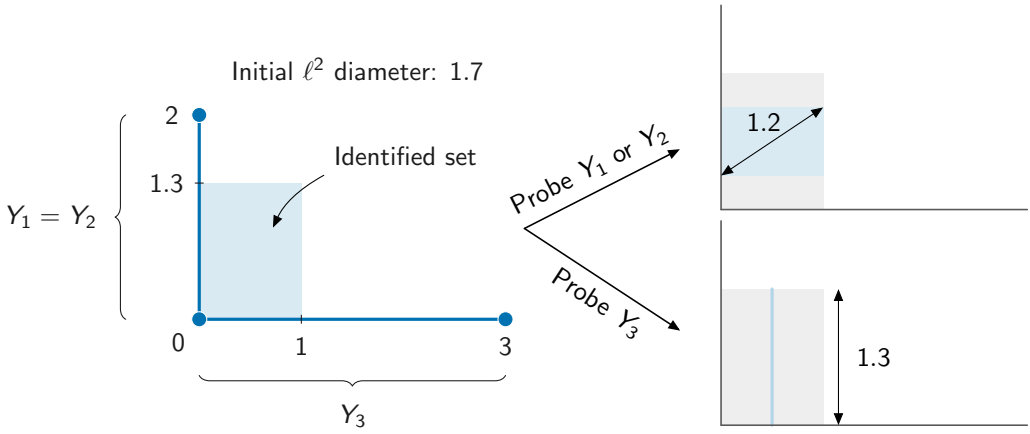
Setup:

- 3 equally weighted observations in $\mathbb{R}^2$
- Outcome sets (line segments):
  - Two vertical segments: origin to $(0, 2)$
  - One horizontal segment: origin to $(3, 0)$
- Probe: corresponding $\mathbf{y}$ revealed exactly

Identified set for $\mathsf{E}[\mathbf{y}]$:

$$[0, 1] \times \left[0, \frac{4}{3}\right]$$

Vector Counterexample



Corollary 1: Lorenz Representation

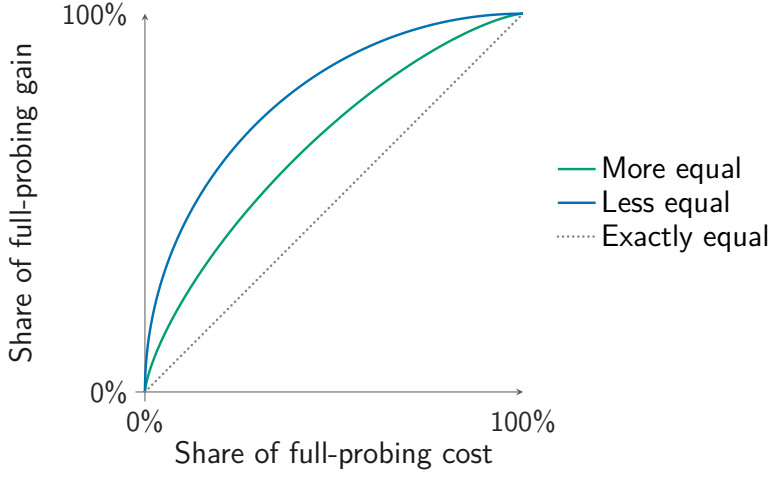
Under Theorem 1, with $\mathrm{E}[v] > 0$: optimal ϕ_b^* at budget b

Lor: Lorenz curve of v/c; weights proportional to cost c

Optimal fraction of full-probing gain:

$$\frac{\mathrm{E}[v\phi_b^*]}{\mathrm{E}[v]} = 1 - \mathrm{Lor}\left(1 - \frac{b}{\mathrm{E}[c]}\right), \quad 0 \leqslant b \leqslant \mathrm{E}[c] < \infty$$

Dual Lorenz Curves



Illustration